

Summary and outline proof of the Perkins Skorokhod embedding

Let

$$\overline{W}_t = \sup_{s \leq t} W_s, \quad \underline{W}_t = \inf_{s \leq t} W_s.$$

Define the lower Perkins boundary by

$$K^*(B) \in \arg \max_{K < B} \frac{C_\mu(B) - C_\mu(K) - K}{B - K}, \quad B > 0,$$

and the upper Perkins boundary by

$$K_*(B) \in \arg \min_{K > B} \frac{C_\mu(B) + B - C_\mu(K)}{K - B}, \quad B < 0.$$

The full Perkins stopping time is then given by

$$\tau^P = \inf \{t > 0 : W_t \leq K^*(\overline{W}_t) \text{ or } W_t \geq K_*(\underline{W}_t)\}.$$

Equivalently,

$$\tau^P = \tau_- \wedge \tau_+,$$

where

$$\tau_- = \inf \{t > 0 : W_t \leq K^*(\overline{W}_t)\}, \quad \tau_+ = \inf \{t > 0 : W_t \geq K_*(\underline{W}_t)\}.$$

Thus $K^*(B)$ is the lower stopping boundary associated with a running maximum equal to B , while $K_*(B)$ is the upper stopping boundary associated with a running minimum equal to B .